

Huzaifah Bin Wasim

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EDUCATION

Master of Science, Computer Science (Machine Learning) (GPA 4.0/4.0)

Georgia Institute of Technology

Aug 2024 – May 2026

Atlanta, GA

- Specialization in Machine Learning.

Bachelor of Electrical Engineering (Conferred)

National University of Sciences and Technology (NUST)

Sep 2018 – Jun 2022

Islamabad, Pakistan

- Elected specialization stream in Artificial Intelligence.

PUBLICATIONS

- A. Wajid, N. Kamal, M. Sharjeel, R. M. Sheikh, H. B. Wasim, M. H. Ali, W. Hussain, F. Zaffar. "A First Look at Private Communications in Video Games using Visual Features." Proceedings on Privacy Enhancing Technologies (PoPETs), 2021(3), pp. 433–452.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Georgia Institute of Technology

Jan 2025 – Aug 2025

Shenzhen, China

- Conducted research on urban mobility optimization and traffic management using machine learning and predictive analytics, contributing to a 15% improvement in prediction accuracy for next-generation smart city simulations.
- Developed and deployed SUMO-based simulation models with advanced statistical analysis and optimization algorithms, reducing average simulation runtime by 30% while maintaining prediction fidelity across traffic scenarios.

Machine Learning Engineer

Adept Technology Solutions

Jan 2024 – Jul 2024

Islamabad, Pakistan

- Developed advanced AI and data science algorithms for ASCEND, a synthetic data generation and anonymization platform, leveraging Python and prompt engineering to optimize LLM-based data augmentation workflows, increasing synthetic data quality scores by 25%.
- Led a full codebase refactor to implement modular, scalable software architecture with CI/CD pipelines and cloud platforms (AWS) using FastAPI for API endpoints, reducing deployment time from hours to under 15 minutes.
- Leveraged PySpark to implement proprietary statistical analysis algorithms and used document processing libraries to preprocess raw data at scale, processing 500K+ documents per pipeline run.

Research Assistant

Robotics and Machine Intelligence Lab (ROMI), NUST

Feb 2020 – Jul 2024

Islamabad, Pakistan

- Co-authored "A First Look at Private Communications in Video Games using Visual Features," published in Proceedings on Privacy Enhancing Technologies (PoPETs) 2021, performing end-to-end ML workflow including data labeling, feature engineering, model training, and hyperparameter tuning.
- Developed the full software stack for a surveillance area coverage optimization project using computer vision, 3D reconstruction, and blind spot identification algorithms, improving area coverage detection by 20%.

Research Intern

TUKL-NCAI Deep Learning Lab

Jun 2020 – Apr 2021

Islamabad, Pakistan

- Conducted research on predictive and generative algorithms using Transformers, LSTMs, GANs, and Deep Q-Networks.
- Applied research to develop, train, and compare short-term residential electrical load forecasting models.

PROJECTS

Parallel Interleave for Maximal End Component Decomposition

C++, Python, Storm, perf, Valgrind

Jan 2026

- Reimplemented the Interleave algorithm for Maximal End Component decomposition in the Storm probabilistic model checker with parallel execution, achieving 2–5x speedup over sequential interleave across the QVBS benchmark suite.
- Built a profiling pipeline using perf and Valgrind with C++ symbol demangling, flamegraph generation, and automated visualization for performance analysis.
- Developed CLI tools for batch profiling (perf record with DWARF call-graphs at 997 Hz, uprobe instrumentation) and automated visualization (flamegraph SVG, filtered text reports).
- Created wrapper functions to resolve C++ template symbol visibility issues that prevented profilers from detecting Interleave entry/exit points.

Legal Knowledge Graph RAG System for Immigration Law

Nov 2025

Python, Neo4j, Graphiti, Gemini, Pydantic, LangChain

- Designed and deployed an end-to-end legal RAG assistant for immigration case law, integrating a Neo4j-based Graphiti knowledge graph containing 264K nodes and 506K relationships with Gemini 2.5 Flash.
- Implemented Pydantic-driven structured extraction to transform unstructured court opinions into a type-safe knowledge graph, attaining a faithfulness score of 0.770 and outperforming NaiveRAG and CBR-RAG by 15–17% with significant reduction in output variance.
- Built a scalable CourtListener ingestion pipeline with rate-limited concurrency and hybrid retrieval, blending vector similarity search with Cypher-based graph traversal to facilitate multi-hop reasoning across precedents and statutes.

PopSign: Generative AI Storybooks for ASL Learning

Oct 2025

Python, Gemini, Sentence-BERT, spaCy, Embedding-Gemma

- Developed a constrained generative AI pipeline with Gemini to author ASL children's stories, using prompt engineering and a custom validation framework to enforce lexical constraints and reduce error rate to 0.
- Created an NLP evaluation system using Sentence-BERT, spaCy, and readability metrics to automatically classify story difficulty across Fountas-Pinnell levels A–Z, enabling data-driven curriculum design.
- Engineered an embedding-based semantic vocabulary sampler with Embedding-Gemma to generate contextually relevant word sets for personalized, level-appropriate story generation.

Surveillance Area Blind Spot Detection

MATLAB, VisualSFM, LIME, Computer Vision

- Led the research project to identify blind spots in a surveillance camera field of view.
- Used computer vision to reconstruct 3D surveilled areas from images and videos.
- Explored explainable AI methods (LIME) to improve the fairness of the model.

SLAM + DQN for TurtleBot3 Navigation

ROS, Ubuntu, TensorFlow, SLAM, Deep Reinforcement Learning

- Combined SLAM with a Deep Q-Network in ROS to improve performance of autonomous TurtleBot3 navigation.

APF + DQN for TurtleBot3 Navigation

ROS, TensorFlow, Python, Gazebo

- Accelerated training of a DQN by integrating an Artificial Potential Field (APF) algorithm for autonomous navigation of TurtleBot3 in Gazebo.

Predictive Object Tracking

Python, Computer Vision, State Estimation

- Implemented a Kalman filter to detect objects and predict their positions to facilitate autonomous vehicle navigation.

Robot Localization

Python, Probabilistic Inference, Particle Filters

- Designed and implemented a Particle Filter with a custom fuzzing algorithm for autonomous vehicle localization in a map.

Robot Path Planning

Python, Search Algorithms, Dynamic Programming

- Implemented A* search and Value Iteration to enable autonomous robot navigation in a warehouse.

PID Controller

Python, Control Systems, Robotics

- Designed and tuned a PID controller to control a quadcopter.

SpaceX Starship Lander

TensorFlow, Unity, OpenAI, RL

- Designed a model of SpaceX's Starship in Unity.
- Designed and deployed a deep reinforcement learning, multi-agent control system using TensorFlow to safely land the rocket on its landing pad.

Smart Garbage Bin

Jetson Nano, Computer Vision, TensorFlow, LIME, RL

- Used transfer learning to retrain MobileNet v2 on a custom garbage dataset.
- Employed OpenAI API to integrate the classifier with a deep RL agent while using LIME to explain the classifier's output.
- Deployed on Jetson Nano.

Fine-tuned French GPT-3 Chatbot

OpenAI API, GPT, Flask, Python

- Developed a custom webpage for GPT-3-based French language response generation. Users could retrain the model on their own dataset.

Sign Language Translation Using OAK-D

Computer Vision, OpenCV, OAK-D, DepthAI

- Developed an OAK-D-based sign language translation tool.

Short-term Residential Load Forecasting

TensorFlow

- Built and tested multiple predictive models based on Transformers, LSTMs, and reinforcement learning for short-term energy consumption predictions.

Task Scheduling in Cloud Computing

Python, Graph Algorithms

- Implemented a topological sorting algorithm for efficient staging and execution of tasks on cloud computing workflows and pipelines.

Optimal Seam Finding

Python, Dynamic Programming

- Implemented an algorithm to find low-cost seams in 2D images to enable resizing without distorting perspective and relative object size.

Finding C and G Rich Regions in DNA Sequences

Python, Divide and Conquer

- Implemented a modified maximum sub-array sum algorithm to locate consecutive CpG nucleotide islands in mammalian DNA.

Scalable RISC-V Multicore Processor

Xilinx Vivado, OpenPiton, N-tiny, FPGAs

- Interfaced a custom-built 32-bit RISC-V (N-tiny) with the OpenPiton framework to build a scalable multicore processor.
- Implemented 4-way 4-word set-associative caches with LRU policy, Avalon-to-Pico interconnect, and AXI-4 Lite interface.
- Deployed on Genesys2 FPGA.

Speech Recognition on Edge

Linux, Kaldi, FPGAs

- Designed and built a custom hybrid speech recognition model using a combination of Hidden Markov Models and Gaussian Mixture Models.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Bash, MATLAB, SQL

ML / DL Frameworks: PyTorch, TensorFlow, Scikit-Learn, Stable-Baselines3, Optuna, Hugging Face, LangChain, Graphiti, Gradio, Streamlit

Reinforcement Learning: OpenAI API, Stable Baselines, Gymnasium, OpenSpiel, Deep Q-Networks

Computer Vision & Robotics: OpenCV, ROS, ROS2, Gazebo, SLAM, Kalman Filters, Particle Filters, OAK-D, DepthAI, LIME, 3D Reconstruction

Generative AI & NLP: LLMs, RAG, Graph RAG, Prompt Engineering, Generative AI, NLP, Sentence-BERT, spaCy, Large Language Model APIs, Model Context Protocol

Specializations: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Data Analysis, AI Safety, Knowledge Graphs, Data Structures, Algorithms

Data & Databases: NumPy, Pandas, SciPy, PySpark, Hadoop, PostgreSQL, MySQL, SQLite, Neo4j, ChromaDB, FAISS, Vector Stores

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD Pipelines, Git, FastAPI, Pydantic

Hardware & Edge: Jetson Nano, FPGAs (Xilinx Vivado, Genesys2), OAK-D Camera

Formal Methods: Storm Model Checker, perf, Valgrind, Probabilistic Model Checking

AWARDS

Finalist, OpenCV AI Competition — Awarded 10 OAK-D cameras for original project idea (Sign Language Translation)

2021